

Exploratory Analysis of Pharmaceutical Data in New Zealand

Data Science Project Report

Course: DATA301
Project Group 19

Jessica Saucedo 300666254

Nam Bui 300665058

Tianyi Xu 300658036

Zion Swinburne 300560705

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Chapter 1

Background and Data

1.1 Background

This project investigates patterns in pharmaceutical use in New Zealand using publicly available summary data from Health New Zealand. The analysis covers initial pharmaceutical dispensings between 2021 and 2025 and focuses on Therapeutic Group Level 2 categories and health districts.

The main objective is to identify differences in the number of patients associated with initial dispensings across therapeutic groups, years, and districts. This provides an exploratory view of how pharmaceutical activity is distributed across New Zealand and how these patterns have changed over time.

Pharmaceutical data are useful for identifying broad patterns in healthcare activity. For example, differences between therapeutic groups may indicate that some classes of medicines are associated with substantially larger numbers of patients than others. Temporal patterns can also indicate whether patient counts are relatively stable, increasing, or declining. District-level comparisons provide an additional geographical perspective.

The analysis is exploratory rather than causal. The observed differences cannot by themselves establish why pharmaceutical patient counts differ between therapeutic groups or districts.

1.2 Research Question

We begin the analysis with the following research question:

- How does utilisation of the top 10 Therapeutic Group Level 2 pharmaceutical categories vary across Aotearoa New Zealand health districts from 2021 to 2025?

The question is addressed through the following supporting questions:

- Which Therapeutic Group Level 2 pharmaceutical categories had the highest patient-associated dispensing counts from 2021 to 2025?
- How did patient-associated utilisation of these categories vary across health districts and over time?
- How did district-level utilisation patterns differ after adjusting for population size?
- What demographic characteristics were associated with differences in population-adjusted utilisation across health districts?

1.3 Data Sources

The primary dataset was obtained from the Health New Zealand Pharmaceutical Data web tool. The Pharmaceutical Data web tool provides summary information about initial dispensings and community pharmaceutical dispensings funded by the New Zealand Government.

The source is:

Health New Zealand. 2026. *Pharmaceutical Data web tool*, version 07 August 2026.

Data extracted from the Pharmaceutical Collection on 09 July 2026.

The pharmaceutical collection contains claim and payment information supplied by pharmacies for subsidised dispensings. The publicly available data are statistical and non-identifiable.

The analysis uses pharmaceutical data covering 2021–2025.

1.4 Data Structure

The main variables used in the analysis are:

- **TherapeuticGrp2**: Therapeutic Group Level 2 pharmaceutical category. This is a categorical variable with multiple therapeutic categories.
- **District**: health district associated with the pharmaceutical record.
- **YearDisp**: year in which the pharmaceutical was dispensed.
- **NumDisps**: number of initial dispensings.
- **NumPpl**: number of patients associated with the dispensing records.

The data therefore contain both categorical and numerical variables. Therapeutic group and district are categorical variables, while year and patient counts are numerical variables.

The analysis uses **NumPpl** rather than **NumDisps** as the primary measure because **NumPpl** represents the number of patients associated with the dispensing records, providing a patient-oriented measure of pharmaceutical activity.

1.5 Census Data

In addition to the pharmaceutical dataset, this project uses a selection of variables from the 2023 New Zealand Census (Stats NZ, 2023), extracted from Stats NZ’s Aotearoa Data Explorer to provide population and demographic context for each health district. Selected variables include **Area**, **Population**, **Age (Life Cycle Groups)**, **Ethnicity**, **Māori Descent Indication**, **Gender**, **Average Number of Children**, **Smoking Status**, **Disability Indication**, **Median Income**, and **Employment Status**.

The Healthy Futures (Pae Ora) Act 2022 merged Capital and Coast DHB and Hutt Valley DHB into one administrative area: Capital, Coast, and Hutt Valley District (Ministry of Health NZ, 2022). In the Health New Zealand dataset, these remain counted as two separate areas, however the census, being completed in 2023, uses the new Pae Ora boundaries. In order to integrate the two datasets, observations for Capital and Coast and Hutt Valley DHBs with

matching `YearDisp` and `TherapeuticGrp2` were combined in the Health New Zealand database before completing a left join on the census data on `District = Area`.

After combining these datasets, we used the district population counts to create two new variables, `Patients_Per_1k` and `Dispensings_Per_1k`; for every 1000 people in a district, what proportion of those each year received an initial dispensing of each Therapeutic Group 2 pharmaceutical.

$$\text{For each observation: } \text{Patients_Per_1k} = \frac{\text{NumPpl}}{\text{population}} \times 1,000$$

Unlike the pharmaceutical dataset, the census data used in this project are not suppressed for small counts, since they are published as complete population statistics rather than service use records.

1.6 Data Quality and Missing Values

The pharmaceutical dataset contains observations where patient counts are reported as “<6”. These values are suppressed because very small counts could potentially risk violating privacy and anonymity of patients who received treatment for rare ailments.

Because “<6” does not provide an exact numerical value, these observations were excluded from analyses requiring exact numerical aggregation. This resulted in the removal of 524 observations (4.23% of the dataset). As the analysis focuses on identifying the top 10 most utilised Therapeutic Group Level 2 pharmaceuticals, we determined that including the suppressed observations would not affect the identification or ranking of the top 10 categories. Therefore, these observations were excluded from the dataset entirely. The potential effect of this exclusion on the broader analysis is considered as a limitation.

The count of suppressed observations for each district was roughly in order and reflective of the area’s population, and the most commonly suppressed Therapeutic Group 2 pharmaceuticals are either used to treat extremely rare afflictions in NZ borders, like `Antimalariants`, pharmaceuticals more easily accessible through non-clinical means, like `Gluten Free Foods`, or generally uncommon pharmaceuticals, like Section 29 medications - eg `Respiratory Stimulants` (Medsafe, 2025).

After removing the suppressed observations, we were able to convert `NumPpl` and `NumDisps` to numerical values.

The dataset was also checked for missing values and duplicate rows before analysis, of which neither were found.

1.7 Data Preparation

The pharmaceutical data were grouped by Therapeutic Group Level 2, year, and district to obtain aggregate patient counts, with `sum(NumPpl)` for each district.

For district-level comparisons, the “New Zealand” national total was removed. This leaves the individual health districts and prevents the national total from being compared directly with individual districts.

The resulting data were used to produce six exploratory visualisations:

1. the top ten Therapeutic Group Level 2 categories by patient count;

2. trends in patient counts for the top ten groups from 2021 to 2025;
3. total patient counts by health district;
4. a heatmap showing the top ten therapeutic groups across districts;
5. a heatmap showing patient rates for the top ten Therapeutic Group Level 2 categories by district;
6. a heatmap showing the top ten demographic variables (positively or negatively) associated with patient rates by district.

Chapter 2

Exploratory Data Analysis

2.1 Top 10 Therapeutic Groups by Patient Count

Figure 2.1 presents the ten Therapeutic Group Level 2 categories with the largest aggregate numbers of patients associated with initial dispensings between 2021 and 2025.

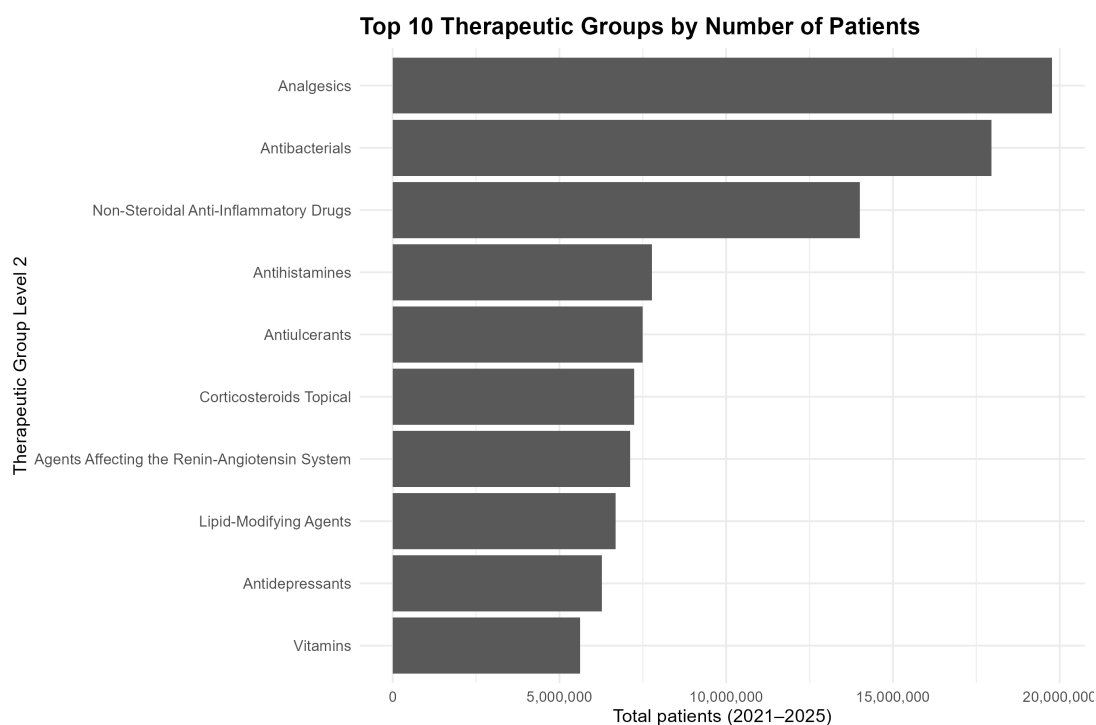


Figure 2.1: Top 10 Therapeutic Groups by Number of Patients, 2021–2025.

Analgesics have the highest aggregate patient count, followed by Antibacterials and Non-Steroidal Anti-Inflammatory Drugs. There is a substantial difference between these three groups and the remaining categories.

The ranking suggests that a relatively small number of therapeutic categories account for approximately 50% of the patient-associated activity represented in the dataset. Analgesics and Antibacterials are clearly separated from the other categories, while Non-Steroidal Anti-Inflammatory Drugs form a second high-volume group.

The remaining categories have considerably smaller totals, although they still involve hun-

dreds of thousands of patient-associated records across the five-year period.

The top ten groups identified were:

1. Analgesics
2. Antibacterials
3. Non-Steroidal Anti-Inflammatory Drugs
4. Antihistamines
5. Antiulcerants
6. Corticosteroids Topical
7. Agents Affecting the Renin-Angiotensin System
8. Lipid-Modifying Agents
9. Antidepressants
10. Vitamins

These categories were subsequently used for the temporal and district-level analyses.

2.2 Changes in Patient Counts Over Time

Figure 2.2 shows the annual patient counts associated with the ten highest-volume therapeutic groups between 2021 and 2025.

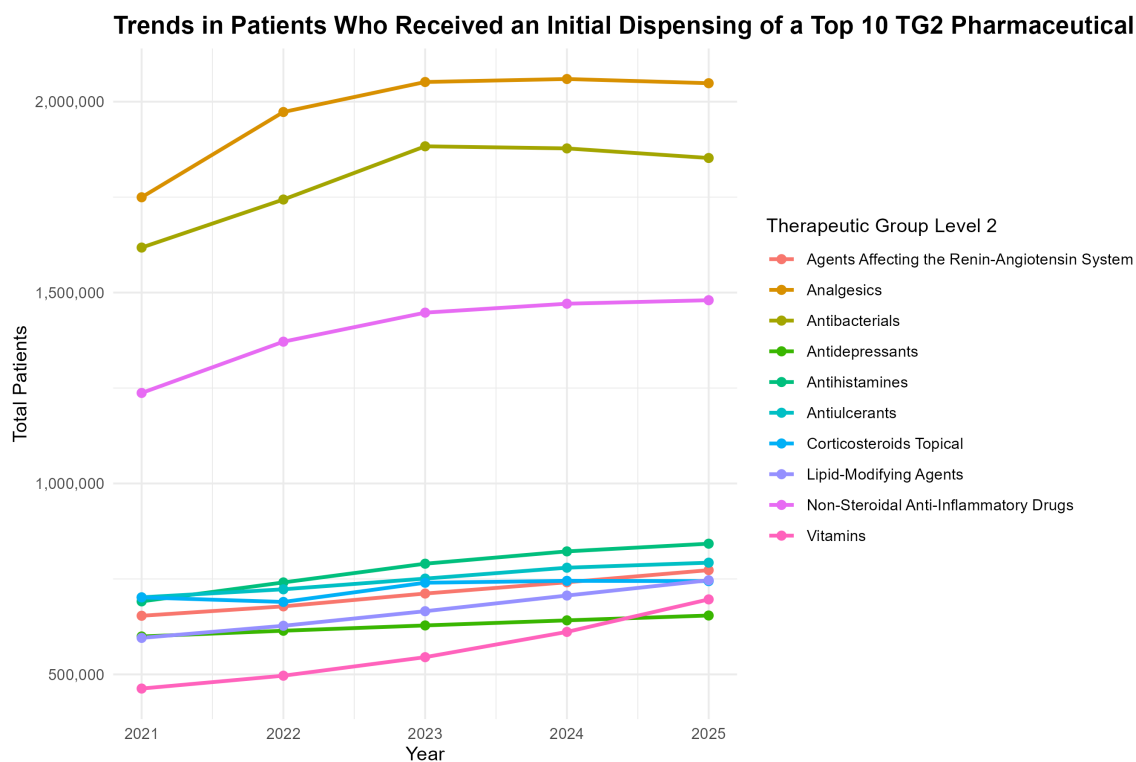


Figure 2.2: Trends in the Number of Patients for the Top 10 Therapeutic Groups, 2021–2025.

Table 2.1: Change in Patient Counts for the Top 10 Therapeutic Groups, 2021–2025.

TherapeuticGrp2	Patient Count (2021)	Patient Count (2025)	Change (Num)	Change (%)
Analgesics	1,749,590	2,048,374	298,784	17.1
Antibacterials	1,617,911	1,852,509	234,598	14.5
Non-Steroidal Anti-Inflammatory Drugs	1,237,133	1,479,989	242,856	19.6
Antihistamines	691,221	842,319	151,098	21.9
Antiulcerants	701,838	792,597	90,759	12.9
Corticosteroids Topical	701,613	744,273	42,660	6.1
Agents Affecting the Renin-Angiotensin System	653,538	773,023	119,485	18.3
Lipid-Modifying Agents	595,608	746,469	150,861	25.3
Antidepressants	599,367	654,226	54,859	9.2
Vitamins	462,889	696,412	233,523	50.4

The temporal patterns are not identical across therapeutic groups. Analgesics remain the largest category throughout the entire period. Their patient count increased substantially between 2021 and 2023, reached a relatively stable level in 2024, and decreased slightly in 2025.

Antibacterials show a similar pattern of strong growth between 2021 and 2023 followed by a gradual decline in 2024 and 2025. The persistence of these groups near the top across multiple years indicates that their high aggregate totals are not driven solely by an odd year’s observations.

Non-Steroidal Anti-Inflammatory Drugs show a more consistent upward pattern across the five years. Their patient count increased from approximately 1.2 million in 2021 to nearly 1.5 million in 2025.

Several of the lower-volume groups also show relatively steady increases. Antihistamines, Antiulcerants, and Lipid-Modifying Agents all increase across much of the study period, however with the lower-volume trends being compressed they difficult to distinguish.

Vitamins show the clearest relative increases among the lower-volume groups. The patient count increases 50% from 2021 to 2025, although its overall level remains below the largest therapeutic groups.

These patterns demonstrate why examining only the five-year total can hide important temporal differences. Two therapeutic groups can have similar aggregate totals while following different trajectories over time.

2.3 District-Level Differences

Figure 2.3 compares the aggregate number of patients associated with initial pharmaceutical dispensings across the 20 individual health districts.

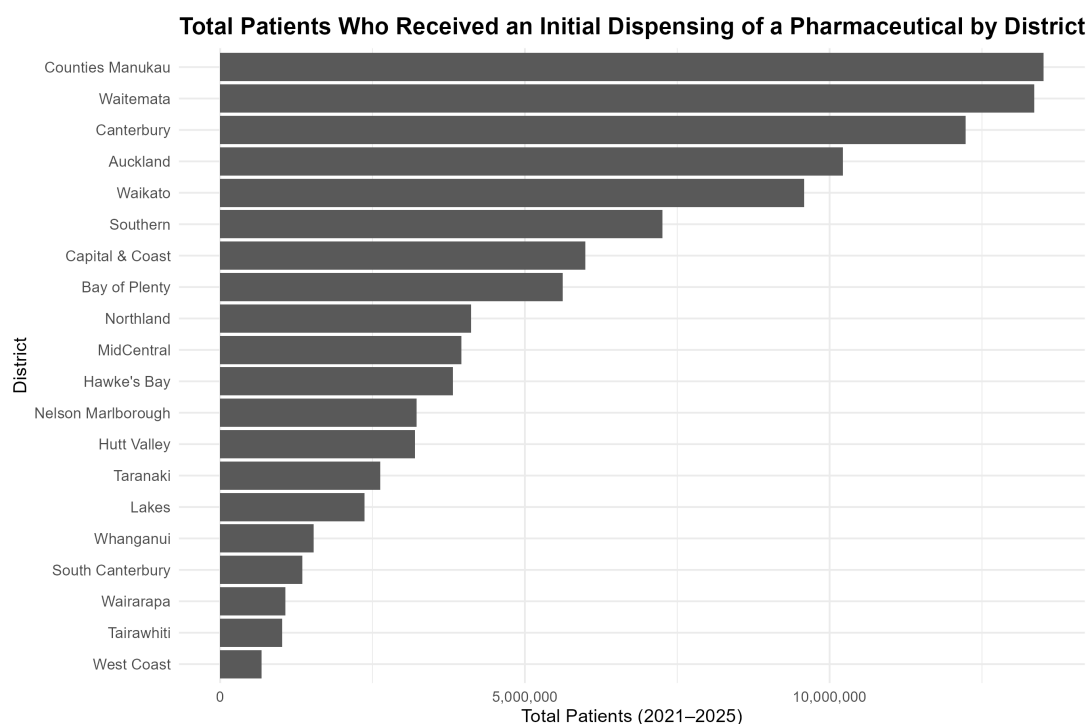


Figure 2.3: Total Number of Patients by District, 2021–2025.

The district comparison shows substantial variation in the aggregate number of patients.

Counties Manukau has the highest total, followed by Waitemata and Canterbury. Auckland and Waikato also have relatively high totals. At the other end of the distribution, West Coast has the lowest aggregate count, followed by Tairāwhiti, Wairarapa, and South Canterbury.

The pattern is strongly associated with the scale of the population represented by each district. Larger districts are expected to have more patients simply because they contain more people.

Consequently, the results should not be interpreted as evidence that residents of one district use pharmaceuticals more frequently than residents of another district.

For example, the difference between Counties Manukau and West Coast is large in absolute terms, but the figure does not account for the different population sizes of the two districts.

A more informative comparison for future analysis would therefore calculate a population-adjusted patient rate, such as the number of patients per 1,000 residents.

2.4 Therapeutic Groups Across Districts

Figure 2.4 presents a heatmap of the top ten therapeutic groups across the 20 districts.

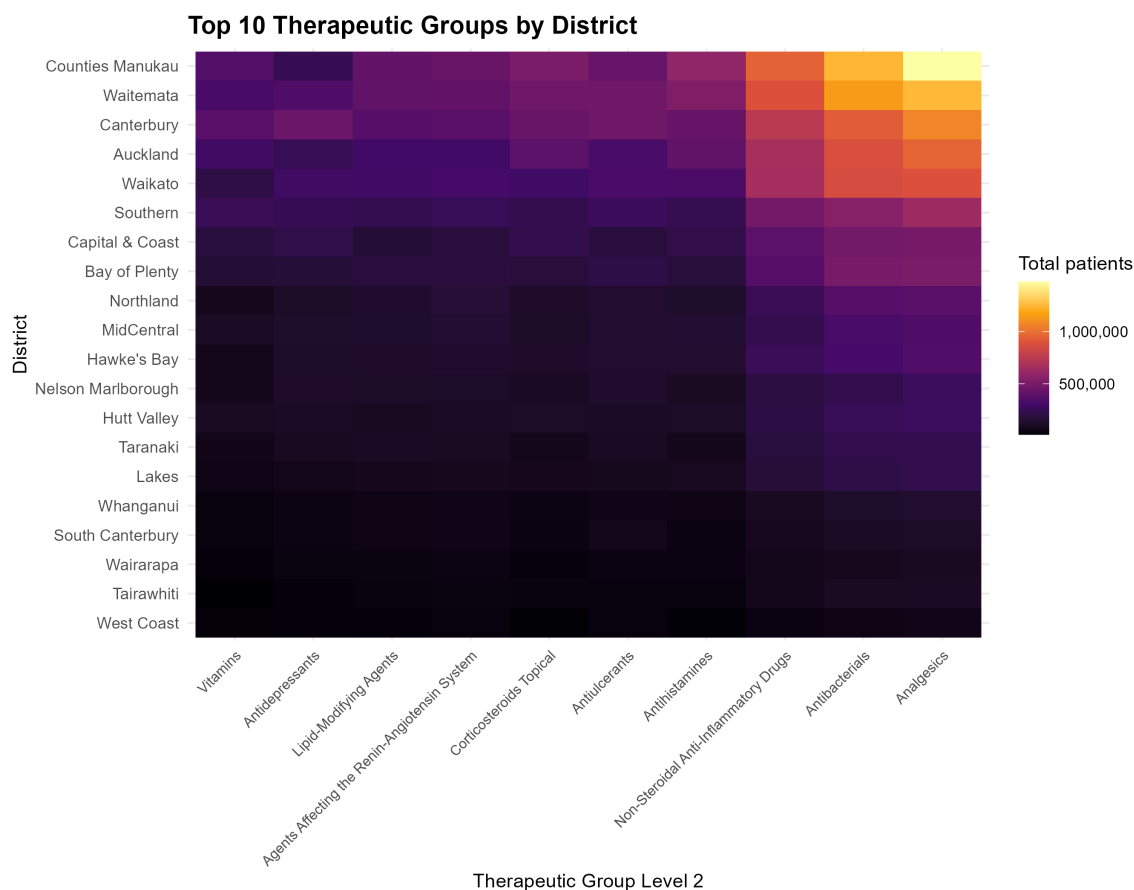


Figure 2.4: Top 10 Therapeutic Groups by Number of Patients Across Districts, 2021–2025.

The heatmap provides a more detailed view than the overall district totals because it considers therapeutic group and district simultaneously, however it must be considered that this is an exploratory visualisation of absolute volume, not relative pharmaceutical use, which will be looked at in a later section.

Analgesics have high patient counts across the major districts, consistent with their position as the highest-volume therapeutic group in Figure 2.1. Antibacterials and Non-Steroidal Anti-Inflammatory Drugs also show relatively high values across the larger districts.

The highest values are concentrated in districts such as Counties Manukau, Waitemata, Canterbury, and Auckland. This pattern is consistent across several therapeutic groups rather than being restricted to a single category.

Smaller districts generally have lower absolute patient counts across most of the therapeutic groups.

This indicates that much of the variation visible in the heatmap is likely related to district size. However, the heatmap can still identify whether particular therapeutic groups behave differently from the general district pattern.

For example, if a district had unusually high values for one therapeutic group relative to its other categories, this could provide a basis for further investigation.

2.5 Comparison of the Main Patterns

The first four figures show that absolute pharmaceutical activity is strongly structured by both therapeutic group and district population size. This gives us cause to make investigate whether rates would be better suited, since raw counts cannot distinguish differences in district size from differences in pharmaceutical activity relative to population

Figure 2.1 identifies which therapeutic groups have the largest aggregate patient counts. Figure 2.2 then shows that these groups do not all follow the same trajectory over time.

Figure 2.3 demonstrates that absolute patient counts differ substantially between districts. Figure 2.4 combines these two dimensions and shows that district differences occur across multiple therapeutic categories.

Taken together, the analysis suggests three main patterns:

1. pharmaceutical patient counts are concentrated in a small number of therapeutic groups;
2. patient counts change differently across therapeutic groups over time;
3. absolute patient counts vary strongly between districts.

These observations provide useful directions for further statistical analysis, particularly analyses that account for population size and examine changes within individual therapeutic groups.

2.6 Population Adjusted Comparisons

The district comparisons in Sections 2.3 showed significant differences in the total patient counts between districts. However, these differences are strongly influenced by population size. To address this patient counts were combined with Census population data to calculate patient rates per 1,000 residents for each therapeutic group and district. It must be stated that the rates for each year, 2021-2025, are calculated based on the district populations of 2023. It is the midpoint of our range, but will not be truly representative of the population each year.

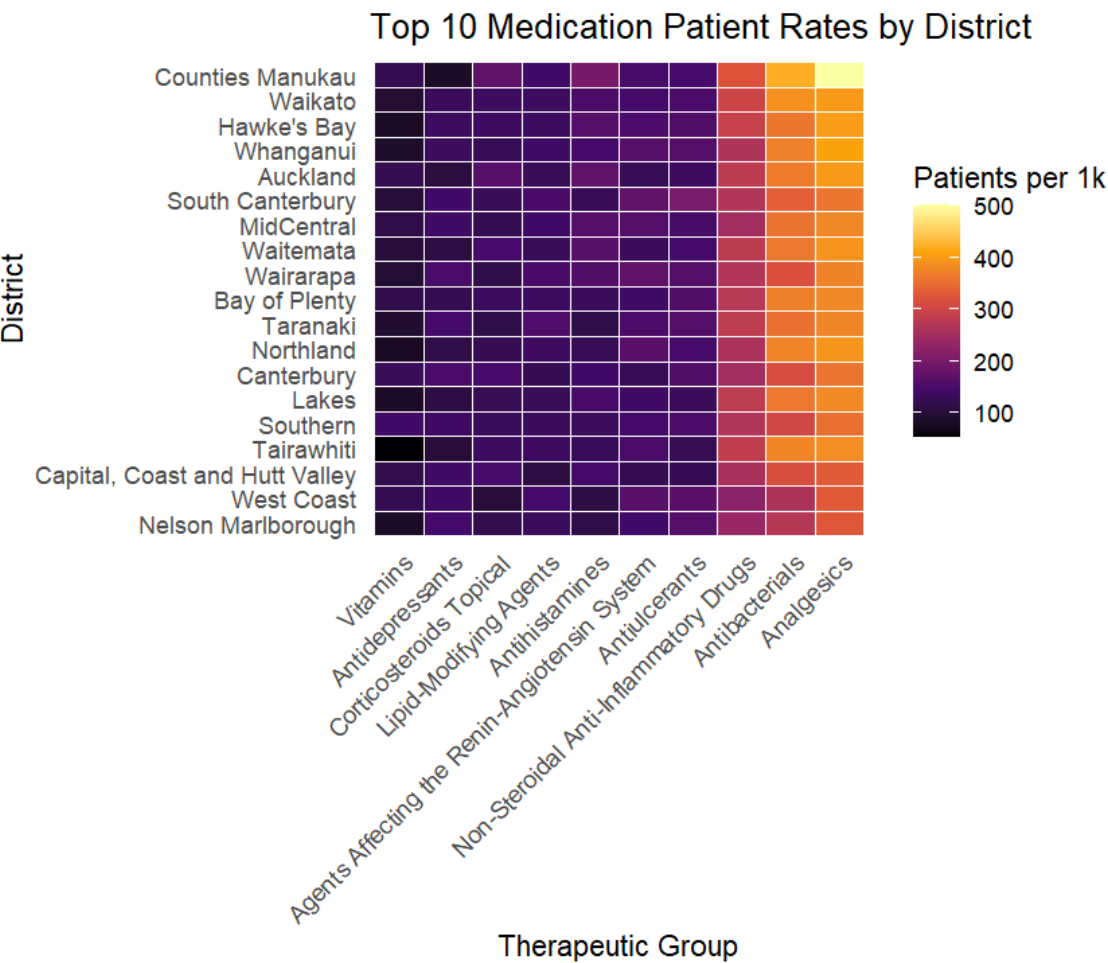


Figure 2.5: Patient Rates for the Top 10 Therapeutic Groups by District.

Counties Manukau had the highest population adjusted patient rate for any therapeutic group, with approximately 500 patients per 1,000 residents for Analgesics, followed by 420 per 1,000 for Antibacterials in the same district. Whanganui (409 per 1,000), Hawke’s Bay (401 per 1,000), and Waikato (398 per 1,000) also showed high Analgesics rates, despite these districts having considerably smaller total patient counts than Counties Manukau.

This adjustment changes the picture considerably compared to the absolute totals in Figure 2.3. Canterbury, which had the third highest total patient count in Table ??, does not appear among the districts with the highest population adjusted rates. In contrast, Hawke’s Bay and Whanganui, which ranked well outside the top five districts by absolute patient count, had the third and fourth highest average patient rates per 1,000 residents. This suggests that population size is an important contributor to differences in the absolute district rankings, which do not necessarily reflect the frequency with which residents receive initial dispensings relative to district population.

To further explore what might explain district-level differences in patient rates, demographic variables from the Census were compared against pharmaceutical patient rates. Figure 2.6 shows the demographic variables most strongly associated with patient rates, ranked by correlation.

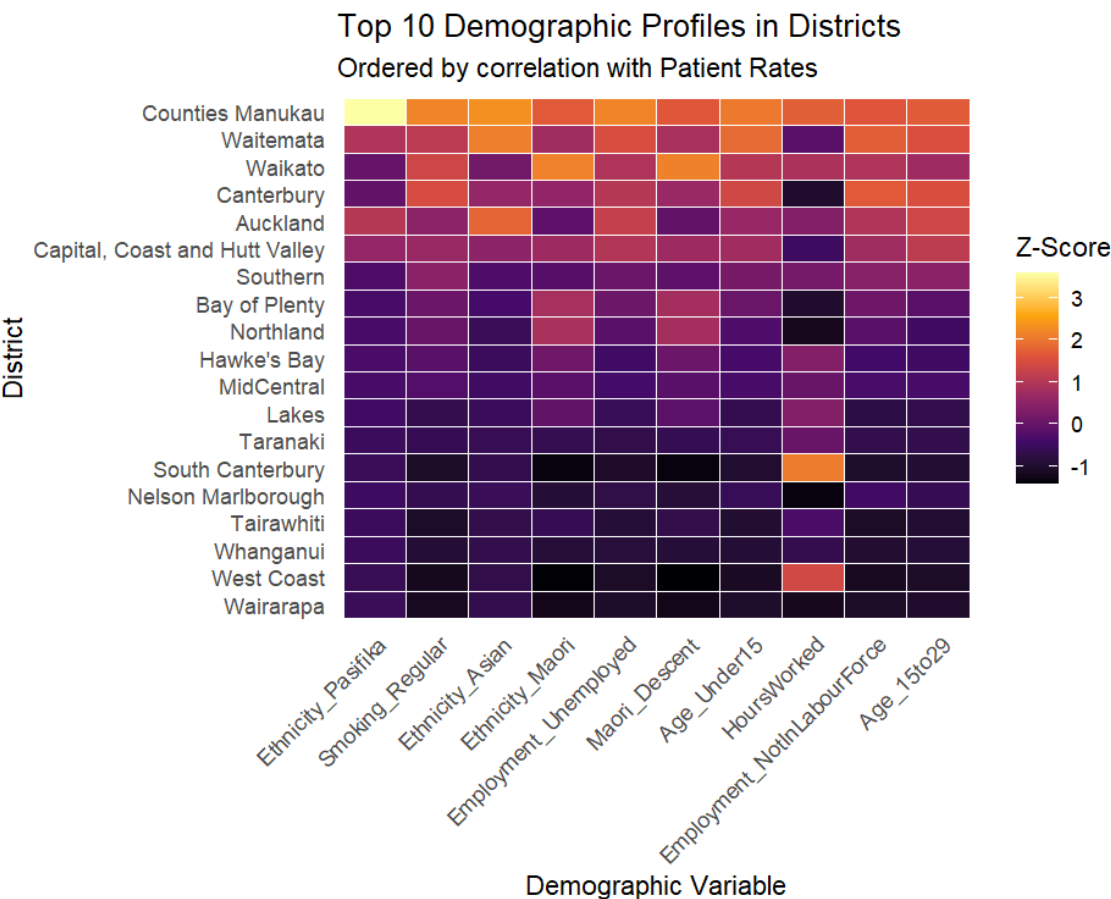


Figure 2.6: Top 10 Demographic Variables Associated with Patient Rates by District.

Ethnicity_Pasifika showed the strongest association with patient rates ($r = 0.65$), followed by Smoking_Regular and Ethnicity_Asian, which were equally associated ($r = 0.50$ each), and Ethnicity_Maori ($r = 0.49$). Employment_Unemployed and Maori_Descent showed moderate associations as well ($r = 0.48$).

These associations are descriptive rather than causal, and should be interpreted with the ethical considerations discussed in Chapter 3 in mind: higher patient rates in districts with particular demographic profiles do not establish that a demographic characteristic causes higher pharmaceutical use, as differences in healthcare access, prescribing practices, or other unmeasured factors could also contribute. Because the analysis uses district-level aggregate data, these relationships are ecological associations and cannot be interpreted as individual-level relationships.

The correlation analysis is based on 20 health districts, so the results should be interpreted cautiously and may be sensitive to individual district, and the demographic variables shown were selected based on the strength of their observed correlations; therefore, the results should be treated as exploratory rather than confirmatory.

Chapter 3

Ethics, Privacy and Security

3.1 Ethical Considerations

The pharmaceutical dataset is derived from an official government health data source and is provided publicly in statistical, non identifiable form. The analysis therefore does not directly involve identifiable individual patients. Nevertheless, pharmaceutical data represent health related information and require careful interpretation, as patterns in pharmaceutical use can be sensitive because they may indirectly reflect the prevalence or treatment of health conditions within populations.

Even though the data is grouped as a summary and do not directly identify individuals, small dispensing counts or combining the data with other datasets could potentially increase the risk of identifying individuals and their health status. Therefore, the analysis will focus on combined patterns at the health district and avoid attempting to identify individuals or making conclusions from very small counts.

Ethical interpretation is also important when comparing dispensing patterns between regions. A high dispensing rate for a particular drug in a given health region does not automatically translate into a greater prevalence of a certain health problem or demographic group. Differences could result from factors such as healthcare access, prescribing practices, or population differences. Similarly, a district with a high absolute patient count may simply have a larger population, and reporting raw totals without this context could create misleading comparisons between communities. Thus, the findings will be reported cautiously and conclusions will be limited to patterns supported by the data.

Māori data sovereignty is pertinent here too since the database has health data from all over Aotearoa New Zealand, including Māori communities. The analysis should therefore avoid deficit based interpretations, unsupported assumptions about Māori health, and generalisations about communities based on dispensing patterns alone.

3.2 Privacy

The public pharmaceutical dataset has already been aggregated and anonymised, and does not contain directly identifiable individual-level information. Small values are suppressed using “<6”, which prevents exact counts from being reported when the number of patients is very small. This suppression is important because combining detailed geographical and pharmaceutical information can increase the risk of identifying small groups. The analysis retains this privacy protection

by treating “<6” as missing rather than attempting to infer or reconstruct the original value.

The census data used in this project also require particular care, and are grouped at the health district (DHB) level rather than at smaller geographic units such as Statistical Area 1 (SA1). More geographically detailed census information, such as SA1, can potentially describe relatively small groups of households: SA1 areas typically cover only 100 to 200 households, with a maximum of 500, and carry a unique publicly available identifying number (Stats NZ, 2026), which could technically make it possible to narrow information down to a small number of households. Although census data are publicly available in aggregated form, combining them with other datasets could increase the level of detail about particular communities. By working with data grouped at the larger health district-level instead, this risk is reduced.

3.3 Data Security

The analysis was carried out using the provided datasets and statistical software. The public datasets themselves do not contain directly identifiable patient information, and project data and results are stored and shared securely, with access limited to project members only and no unnecessary sharing of potentially sensitive health information. Since both datasets are publicly available and already summarised in groups, no additional access controls were required beyond this, and the project did not attempt to access restricted patient-level information.

If more detailed or non public health data were used in future work, such as SA1 level data or individual patient records, stronger security measures would be required. These could include restricted access, encrypted storage, controlled sharing of datasets, and removal of unnecessary identifying variables.

Chapter 4

Key Findings and Discussion

4.1 Main Findings

The exploratory analysis identified several important patterns.

First, Analgesics have the largest aggregate number of patients associated with initial dispensings between 2021 and 2025. Antibacterials and Non-Steroidal Anti Inflammatory Drugs are the next largest groups.

Second, the top therapeutic groups do not all follow the same temporal pattern. Analgesics and Antibacterials increase strongly in the early part of the study period before becoming more stable or declining slightly. Other categories, such as Non-Steroidal Anti Inflammatory Drugs, show a more consistent upward trend.

Third, there are large differences in absolute patient counts between districts. Counties Manukau, Waitemata, and Canterbury have the largest totals, while West Coast has the smallest.

Fourth, the district heatmap indicates that the district-level differences occur across multiple therapeutic groups. This suggests that overall district size is an important factor in explaining the absolute differences.

Fifth, when patient counts are adjusted for population size, the ranking of districts changes considerably. Counties Manukau remains the highest, but districts such as Hawke's Bay and Whanganui, which did not appear among the top districts by absolute count, show some of the highest patient rates per 1,000 residents, as discussed in Section 2.6.

4.2 Interpretation of Patient Counts

An important limitation of the analysis is the interpretation of `NumPp1` after aggregation.

The variable represents the number of patients associated with pharmaceutical records. When values are summed across multiple therapeutic groups or years, the resulting total should not be interpreted as the number of unique individuals in New Zealand.

For example, the same person may be associated with multiple therapeutic groups or may appear in more than one year. Therefore, the aggregate values represent patient counts associated with the relevant records rather than a deduplicated population count.

This distinction is particularly important when comparing the five year totals in Figures 2.1 and 2.3.

4.3 Population Size and District Comparisons

The district analysis highlights the importance of using appropriate denominators.

Absolute patient counts are useful for understanding the overall volume of pharmaceutical activity, but they are not sufficient for comparing pharmaceutical use between populations of different sizes.

For example, Counties Manukau has a much larger aggregate patient count than West Coast. This does not mean that an individual living in Counties Manukau is necessarily more likely to receive an initial dispensing.

A population adjusted measure, calculated in Section 2.6, allows the analysis to distinguish between differences caused by population size and differences in the rate of pharmaceutical activity.

The Census dataset was used for this type of extension. Population variables were integrated with district-level pharmaceutical data to calculate rates per population, and demographic variables were also examined for their association with these rates.

4.4 Potential Further Analysis

The exploratory results suggest several directions for further analysis.

First, the temporal trends could be analysed using year over year percentage changes rather than absolute patient counts.

Second, the demographic associations identified in Section 2.6 could be investigated further, for example by examining whether differences in patient rates by ethnicity or employment status remain after accounting for other demographic factors simultaneously.

Finally, statistical models could be used to determine whether observed differences between districts remain after controlling for population size and other explanatory variables.

4.5 Limitations

Several limitations should be considered.

1. The suppression of small counts: values reported as “<6” cannot be converted into exact numerical observations and were therefore excluded from numerical aggregation.
2. Analysis uses aggregate patient counts rather than unique patient identifiers. The results therefore describe patient associated records and cannot be interpreted as the number of distinct people represented across the entire five-year period.
3. Although population-adjusted rates were calculated in Section 2.6, absolute district totals are still strongly influenced by population size and should be interpreted with this in mind throughout the report.
4. Population-adjusted rates use 2023 Census population data as the denominator, which may not exactly represent district populations during each year of the 2021–2025 pharmaceutical dataset.

5. The pharmaceutical and Census datasets use different administrative structures, requiring Capital & Coast and Hutt Valley records to be combined before joining. This introduces a potential source of aggregation or boundary mismatch.
6. The analysis is descriptive: the observed patterns do not demonstrate causal relationships.
7. Dispensing patterns may reflect differences in healthcare access, prescribing practices and treatment pathways in addition to differences in underlying health needs.
8. The analysis focuses on Therapeutic Group Level 2 categories. Differences between individual medicines within a therapeutic group are therefore not examined.

Chapter 5

Conclusion

This project explored pharmaceutical patient patterns in New Zealand between 2021 and 2025 using Health New Zealand pharmaceutical data.

Analgesics had the highest aggregate number of patients associated with initial dispensings, followed by Antibacterials and Non-Steroidal Anti Inflammatory Drugs. The temporal analysis showed that therapeutic groups followed different trajectories, with some increasing steadily while others increased initially and then became more stable.

There were also substantial differences in absolute patient counts between health districts. Counties Manukau, Waitemata, and Canterbury had the largest totals, while West Coast had the smallest.

The results demonstrate that pharmaceutical activity varies across therapeutic groups, time, and geography. However, absolute patient counts are influenced by population size and should not be interpreted as individual level pharmaceutical usage rates.

Population-adjusted rates, calculated by integrating Census data with the pharmaceutical dataset, showed that the ranking of districts changes considerably once population size is accounted for, with districts such as Hawke's Bay and Whanganui showing notably high patient rates despite modest absolute totals. Demographic variables from the Census, particularly ethnicity and smoking status, were also associated with patient rates across districts, although these associations are descriptive rather than causal. Overall, the analysis demonstrates that pharmaceutical dispensing patterns cannot be meaningfully compared across districts using absolute counts alone, and that population size and the aggregate nature of the data must be considered when interpreting geographical and demographic differences.

Appendix A

Additional Results

A.1 Data Preparation Checks

The pharmaceutical dataset was inspected before analysis to identify its structure, variable types, missing values, and duplicated observations.

The patient count variable contained observations reported as “<6”. These values were identified before numerical conversion and treated as missing.

The national “New Zealand” total was removed from the district-level analysis so that the remaining observations represented individual districts.

The resulting dataset was then grouped by therapeutic group, year, and district to produce the aggregate values used in the figures.

A.2 Summary of the Four Main Figures

Table A.1: Summary of the Main Exploratory Analysis Figures

Figure	Analysis	Purpose
Figure 1	Top 10 Therapeutic Groups	Identifies the therapeutic groups with the highest aggregate patient counts.
Figure 2	Patient Trends	Examines changes in patient counts from 2021 to 2025.
Figure 3	District Comparison	Compares aggregate patient counts across health districts.
Figure 4	District Heatmap	Examines therapeutic group patterns across health districts.

Appendix B

Individual Contributions

B.1 Group Contributions

- **Jessica Saucedo:** Bibliography management.
- **Nam Bui:** Repository management and project organisation.
- **Tianyi Xu:** Report and LaTeX document preparation.
- **Zion Swinburne:** Data collection and Census data collection.

Appendix C

References

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